DISEIL: Demonstration Distillation for Sample-Efficient Imitation Learning

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Abstract

In interactive imitation learning, a learner practises a task and an expert takes over to demonstrate the right actions. Every takeover costs expert time, which compute cannot reduce. Under a fixed budget, what each demonstration contains decides the result. Existing methods decide only when to take over. They call the expert at the first step of an episode where the policy becomes uncertain. An episode that runs from start to end without achieving the task is a failed episode. Two decisions are then left to chance: which failed episode to correct, and where the demonstration begins. We present **DISEIL** (**D**emonstration d**I**stillation for **S**ample-**E**fficient **I**mitation **L**earning), which decides both. After each round of practice, DISEIL groups the failed episodes into modes of failure. For each mode it prescribes where a new demonstration should begin, and checks the start is reachable before any expert time is spent. We evaluate on five tasks, with image and state observations, ten settings in all, at a budget of 20 demonstrations, against the best baselines. DISEIL reaches the highest mean success rate in all ten, on average about 3 percentage points above the strongest baseline.

Introduction

Behaviour cloning fits a policy to an expert's actions on the states the expert visited (Pomerleau 1988; Bain and Sammut 1995). The learner's own small errors then carry it into states it never saw, where errors compound. That drift is called covariate shift (Shimodaira 2000; Ross, Gordon, and Bagnell 2011). Interactive imitation learning stops the drift: the learner acts, and the expert corrects it on the states the learner actually reaches (Ross, Gordon, and Bagnell 2011; Celemin et al. 2022). The price is expert effort, the cost that binds (Mandlekar et al. 2018; Khazatsky et al. 2024). Policy performance scales with how much of the task the demonstrations cover, and extra demonstrations inside a covered region add little (Lin et al. 2024). So once an allowance of B demonstrations is fixed, what each one contains is all that is left to choose (Figure 1).

The DAgger family calls the expert only when a test on the current state fires. Methods differ only in what that test measures. Some measure the gap between the learner's action and the expert's (Zhang and Cho 2017). Others measure the spread of predictions under dropout (Menda, Driggs-Campbell, and Kochenderfer 2017) or across an ensemble (Menda, Driggs-Campbell, and Kochenderfer 2019). Others

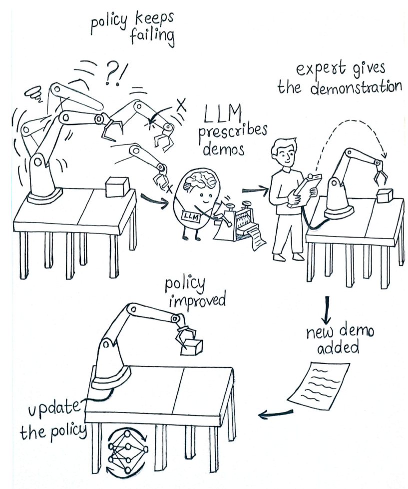

Figure 1: A round of interactive imitation learning ends with a set of failures and one demonstration to spend on it. Existing methods decide only when to ask. DISEIL also decides which failure the call is spent on, and which configuration the expert starts from.

use novelty and risk (Hoque et al. 2021a), or a diffusion policy's own per-step denoising loss (Lee, Kang, and Kuo 2025; Chi et al. 2023). Each reduces the state to one number and uses it the same way. Asking for a demonstration takes three decisions. When to call the expert. Which of the round's failed attempts to spend the call on. Where the corrective demonstration begins. Every method above decides when. Which attempt and where to begin are then settled by default. The attempt corrected is whichever one fired the test first. The demonstration begins in the state that attempt was already in. Under an unbounded budget those two defaults cost nothing, because every mode is corrected eventually. Under a fixed budget they are the whole problem.

One number attached to one state cannot say whether two failures are the same mistake or two different ones. So a mode that recurs often is corrected often, while a rare mode waits. The demonstration also starts at the state that fired the test, frequently one the policy has already ruined. The expert then shows how to recover from the damage, not how to avoid it. The failures within a round form a small set with structure, and geometry recovers it. We describe the configuration of the robot and the objects at the step where the policy first becomes uncertain. Failures grouped by that description separate into geometrically distinct modes. Whether those modes are worth allocating a budget over is an empirical question, and our ablations answer it.

Demonstration distillation for sample-efficient imitation learning (DISEIL) keeps the existing test for when to call the expert. It then decides which failure to correct and where the demonstration begins. Each round, DISEIL anchors every failed attempt at the step where its loss first crosses the threshold, not at the step of highest loss. It groups the failures by the configuration at that anchor step. Among the modes close to the largest in size, it targets the one with the highest mean peak loss. For that mode it prescribes a starting configuration not yet demonstrated. The prescribed start is checked against a constraint graph of what the task permits, before any expert time is spent. A round of failures is distilled into one demonstration that stands for a mode, not for a single episode.

We contribute three things. First, we separate the rule for asking into three decisions: when, which and where. The DAgger family answers only when. Second, we present DISEIL. It reaches the highest mean success rate on unseen test episodes in all ten settings, on average 2.80 percentage points above the strongest baseline in each. Third, we locate the advantage in how the budget is allocated, not in how informative each demonstration is on its own. Removing the mode structure costs a mean 4.37 points of success while information gain does not fall, the signature of a redundantly spread budget.

Related Work

Interactive imitation learning. Dataset aggregation corrects the covariate shift of behaviour cloning (Shimodaira 2000). The expert labels the states the learner visits, and the policy is refit on all data so far. The error bound then grows linearly with episode length, not quadratically (Ross and Bagnell 2010; Ross, Gordon, and Bagnell 2011). Every method compared here uses it. Intervention-weighted methods reweight the data once collected (Mandlekar et al. 2020; Liu et al. 2023b). They change how a round's data is fitted, not which data it buys.

Deciding when to call the expert. Labelling every visited state is expensive, so query-efficient variants add a query gate: a test that scores each visited state and hands control to the expert at the first score above a threshold. The five gated methods here differ only in that score. SafeDAgger predicts deviation from the expert with a classifier (Zhang and Cho 2017). DropoutDAgger measures how tightly dropout samples concentrate about the expert's action (Menda, Driggs-Campbell, and Kochenderfer 2017). EnsembleDAgger combines policy-ensemble variance with an action discrepancy (Menda, Driggs-Campbell, and Kochenderfer 2019). ThriftyDAgger adds a novelty term to a task-risk estimate (Hoque et al. 2021a). Diff-DAgger reads a diffusion policy's per-step denoising loss (Chi et al. 2023) as an uncertainty signal (Lee, Kang, and Kuo 2025). DISEIL uses that same loss to locate a failure and to measure information gain. Human-gated variants leave the decision to the operator (Kelly et al. 2019). Others change only the rule for handing control back (Hoque et al. 2021b). Every gate answers when to call the expert. None says which failed episode to correct, or where the demonstration should begin.

Demonstration selection. Active learning ranks a pool of candidates by an acquisition function (Settles 2009), usually uncertainty (Houlsby et al. 2011) or coverage. Core-set gets coverage from a greedy k-centre rule (Sener and Savarese 2018), BADGE from k-means++ over gradient embeddings (Ash et al. 2020). Sub-trajectory retrieval ranks segments of a demonstration corpus (Memmel et al. 2025). Every method above ranks an existing pool. A round's failed episodes are such a pool. One knockout in the Ablations (A3) ranks them by uncertainty alone: the active-learning rule with its coverage term removed. A coverage rule over the same failures is our grouping's nearest competitor, not run here. No ranking method can name a starting configuration the pool does not contain. DISEIL names one. Dataset distillation compresses an already collected training set into a small synthetic one (Cazenavette et al. 2022). It runs after collection; demonstration distillation runs before it.

Language models and standard components. Language models have been used in robotics as planners, code generators and reward designers (Ahn et al. 2022; Liang et al. 2023; Ma et al. 2024). Given video frames and a record of what the robot did, a model can name a failure's cause well enough to drive recovery (Liu, Bahety, and Song 2023; Duan et al. 2025). The same models are unreliable at judging distance and spatial relations from pixels (Chen et al. 2024a; Fu et al. 2024). DISEIL therefore hands the model the geometry and a constraint graph of what the environment permits, and asks only for a cause and a request. No model here emits an action. Letting a model propose to an external solver is established (Liu et al. 2023a), as is feeding a checker's verdict back for a second proposal (Chen et al. 2024b). Here the checked object is a demonstration not yet collected, not a plan about to be executed. The rest of the machinery is standard and is not claimed: the clustering, the cluster count, the diversity rule and the grid path checker. The Method cites a source for each, where it is used.

Problem Formulation

The policy is any function f_θ that maps an observation to an action. It must also report a per-step loss at each state-action pair it visits,

$$\ell_t^{(i)} \;=\; \mathcal{L}\big(f_\theta,\; s_t^{(i)},\; a_t^{(i)}\big), \tag{1}$$

where i indexes an episode and $a_t^{(i)}$ is the action the policy itself executed. The requirement stops there. For a diffusion policy $\mathcal{L}$ is the denoising loss (Chi et al. 2023); for a discrete policy it is the negative log-likelihood of the action.

The policy is fitted to an initial set $\mathcal{D}_0$ by behaviour cloning (Pomerleau 1988; Bain and Sammut 1995). Round r opens by running the policy on a fresh pool drawn from the task's reset distribution. The episodes it fails form the round's failure set,

$$\mathcal{F}_r = \{\, f_i = (\tau_i,\ \ell^{(i)}_{1:T_i}) \ :\ \tau_i \text{ is a failed episode of } f_\theta \,\}. \tag{2}$$

The count $N = |\mathcal{F}_r|$ varies by round, and T_i is the length of episode τ_i. The round closes by acquiring D demonstrations from the expert $\pi^\star$ and adding them to the training set (Ross, Gordon, and Bagnell 2011),

$$\begin{aligned} &\mathcal{D}_r = \mathcal{D}_{r-1} \cup \{d_{r,1}, \ldots, d_{r,D}\}, \\ &\theta \leftarrow \arg\min_\theta\ \mathbb{E}_{(s,a)\sim\mathcal{D}_r}\big[\mathcal{L}_{\mathrm{BC}}(f_\theta, s, a)\big] \ \text{ for } r \equiv 0 \pmod m. \end{aligned} \tag{3}$$

The loop halts when $|\mathcal{D}_r| - |\mathcal{D}_0| = B$, so it runs B/D rounds. Retraining runs at a per-task cadence m set identically for every method. Success is measured on a fixed set of test episodes, shared by every method and seen by none while demonstrations are collected.

Every method here spends a round by the same three-part rule, $A = (t^\star, C_{\mathrm{tgt}}, \xi)$. The step $t^\star$ is where the expert takes over: the *when*. The mode C_{tgt} is the failure the round is spent on: the *which*. The state ξ is where the corrective demonstration begins: the *where*. A query gate computes $t^\star$ alone, reading C_{tgt} and ξ off whichever failed episode tripped its threshold. The uniform-random control sets all three by sampling. DISEIL keeps that same $t^\star$ and computes C_{tgt} and ξ itself. Nothing else in the loop changes, so the comparison is controlled.

DISEIL

DISEIL runs four stages once per round (Figure 2). It perceives the round's failures and partitions them into failure modes. It then prioritises one mode and prescribes the D demonstrations the expert is asked to supply. Two models fill three roles. The *vision-language model* (Bai et al. 2025) reads rendered frames. The *prescription model* (PM) (Yang et al. 2025) is a text-only reasoning model that labels each failure's root cause and prescribes the round's request. Neither model enters the partition.

Perceive. A round opens by describing each failure twice, from the same step of the same episode. The geometric description feeds the partition. The language one feeds the prescription. The two never mix. Each failure is anchored at one step: the first at which the loss crosses a threshold η and stays above it for K consecutive steps,

$$t_i^\star = \min\{\, t : \ell_u^{(i)} > \eta \ \forall u \in [\,t,\, t+K-1\,] \,\}. \tag{4}$$

If the loss never crosses η, the anchor falls back to $\arg\max_t \ell_t^{(i)}$. The threshold is a quantile of the training-loss distribution, recalibrated at every retrain. Equation 4 is Diff-DAgger's rule, used unchanged (Lee, Kang, and Kuo 2025). The highest-loss step arrives late, so an expert taking over there inherits a corrupted state with little episode left.

Each failure is then reduced to a six-dimensional vector, read from the simulator's internal state at step $t_i^\star$. On the manipulation tasks it is

$$\begin{aligned} &\phi_i = \big[\, p_{x,i},\ p_{y,i},\ \sin\psi_i,\ \cos\psi_i,\ \rho_i,\ \delta_i \,\big], \\ &\rho_i = t_i^\star / T_i, \qquad \delta_i = \big\| p_i^{\mathrm{ee}} - p_i^{\mathrm{obj}} \big\|_2. \end{aligned} \tag{5}$$

Here $(p_{x,i}, p_{y,i})$ is the planar position of the task-relevant object and ψ_i is its yaw. The yaw enters through its sine and cosine, so two yaws either side of $\pm\pi$ are not falsely far apart. The fraction of the episode completed before step $t_i^\star$ is ρ_i, and δ_i is the end-effector-to-object distance. Each task refills the same six slots. The descriptor is small because the failure sets are small: over a few dozen points, weakly informative dimensions flatten the pairwise distances that decide which clusters merge. The supplementary measures the effect.

In parallel, three rendered frames go to the vision-language model: one at the episode's start, one at step $t_i^\star$, one at its end. The model returns a short spatial account. The prescription model turns that account into a root cause and a trajectory phase. Both come from a fixed vocabulary held in the constraint graph $\mathcal{K}$ of the Prescribe stage below, not invented. The division of labour follows the evidence: these models name a cause competently when handed structured evidence (Duan et al. 2025; Liu, Bahety, and Song 2023), but reason poorly about metric geometry from pixels alone (Chen et al. 2024a; Fu et al. 2024).

Partition. A generic clustering step $\mathcal{C}$ groups the round's failures, running over the standardised descriptors,

$$\begin{aligned} &\tilde{X}_i = \frac{\phi_i - \mu_\phi}{\sigma_\phi}, \qquad \{C_1, \ldots, C_{k^\star}\} = \mathcal{C}\big(\tilde{X},\, k^\star\big), \\ &k^\star = \arg\max_{k \in [2,\, k_{\max}]} \mathrm{sil}(k). \end{aligned} \tag{6}$$

Here μ_ϕ and σ_ϕ are the per-dimension mean and standard deviation of the round's own descriptors $\{\phi_i\}_{i=1}^N$, and the cluster count is capped at $k_{\max} = \max(2, \min(6, N-1))$. The clustering step is generic by design. Agglomerative clustering is the choice made here (Ward 1963) and k-means would serve (Lloyd 1982). The silhouette criterion picks $k^\star$, used unmodified (Rousseeuw 1987; Pedregosa et al. 2011). DISEIL claims a partition step, not one implementation of it.

Each mode carries its centroid in raw pose coordinates; its mean peak loss $\bar{L}_C$, averaged over its members' $\mathrm{peak}_i = \max_t \ell_t^{(i)}$; and a representative $\mathrm{rep}(C)$, the member nearest the cluster mean in standardised space. The dominant mode $C^\star$ holds the most members, ties broken by mean peak loss. The peak scores a failure while the first crossing locates it. Below four remaining failures each failure becomes its own single-member mode, and Equation 7 reduces to selecting the failure of highest peak loss. A mode is named by the majority root cause among its members, since reporting a failure in mode 2 says nothing.

Prioritise. Prioritise picks the target mode C_{tgt}. The Prescribe stage below picks the starting state ξ. Consider every mode whose size is within one member of the largest. The round goes to whichever of them has the highest mean peak loss,

$$C_{\mathrm{tgt}} \ = \ \arg\max_{C\,:\,|C| \geq |C^\star|-1} \bar{L}_C. \tag{7}$$

The size constraint keeps the target inside the bulk of the round's failures, so a mode that barely exists cannot capture the budget on one badly failed episode. Because one

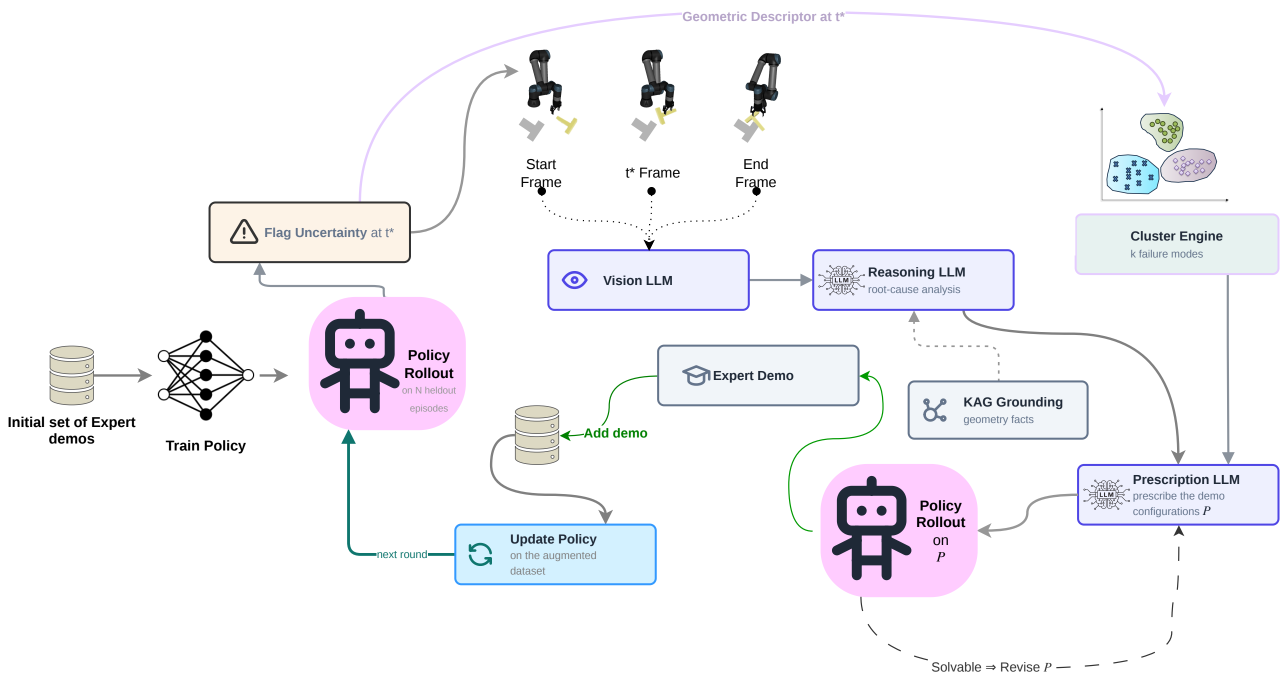

Figure 2: DISEIL. Equation 4 returns the step $t^\star$ at which the policy first becomes uncertain. A query gate computes that step and stops. DISEIL continues from it. The perception branch meets the geometric branch only at the prescription, so no model output enters the partition. "Reasoning LLM" and "Prescription LLM" are the two roles of the prescription model, a large language model. "KAG Grounding" is the knowledge-augmented-generation constraint graph $\mathcal{K}$. The supplementary specifies the solvability screen, drawn here as the rollout and revise branch, where the policy is run once from the prescribed start.

demonstration rarely removes a mode outright, a rule of this shape can return the same mode round after round. DISEIL therefore has a cluster memory, which discounts modes already corrected. The memory is a configurable task-specific component outside the core rule, on in every run here. The prescription model is then shown at most κ cited failures from the target mode: the representative first, then the worst-loss failure, then farthest-point selection (Eldar et al. 1997) to span the mode.

Prescribe. The prescription model receives three inputs: the target mode's anchor geometry, the cited failures with their root-cause labels, and the task's constraints. It returns the round's request, with a confidence score and a rationale. Each requested demonstration takes one of two forms. A *targeted correction* names one cited failure; that exact episode is re-instantiated and the expert takes over at the anchored step. A *bridging placement* names two or three cited failures and asks for a new configuration between them, from which the expert demonstrates a complete episode. Bridging lets a prescription be easier than any failure it addresses. Suppose a mode lies far outside what the policy can solve. A targeted correction there asks for a large jump; a bridged one asks for a step the policy can absorb. The graded-placement argument is reverse curriculum generation's (Florensa et al. 2017), applied to a demonstration request rather than a training reset.

A model asked for a configuration of the world will sometimes name one the world cannot produce. The constraint graph $\mathcal{K}$ makes such a request checkable. It holds workspace bounds, spawn ranges, reachability, controller limits, the success predicate and the failure-mode vocabulary as structured knowledge. It is a robot knowledge base (Tenorth and Beetz 2013), not a retrieval index over text (Lewis et al. 2020; Edge et al. 2024). Verification is a loop. The model proposes a configuration. A map g turns it into a reset specification ξ. A test V checks ξ. If ξ fails, the violation returns as feedback and the model proposes again,

$$\begin{aligned}
\mathrm{cmd}^{(j)} &= \mathrm{PM}\big(C_{\mathrm{tgt}},\ S,\ \mathcal{K},\ \mathrm{violation}(\xi^{(j-1)})\big),\\
\xi^{(j)} &= g\big(\mathrm{cmd}^{(j)}\big),\\
V(\xi) &= \mathbf{1}\big[\xi \in \mathcal{W}_{\mathcal{K}}\big] \wedge \mathbf{1}\big[\mathrm{reachable}_{\mathcal{K}}(\xi)\big]\\
&\quad \wedge\ \mathbf{1}\big[\mathrm{valid\text{-}path}_{\mathcal{K}}(\xi)\big].
\end{aligned} \tag{8}$$

The workspace bounds are $\mathcal{W}_{\mathcal{K}}$, and the conjuncts of V are the constraints $\mathcal{K}$ holds for that task. There is no feedback on the first attempt, so $\mathrm{violation}(\xi^{(0)}) = \emptyset$. On the grid task the last conjunct instead checks that a path exists, using $A^\star$ and breadth-first search (Hart, Nilsson, and Raphael 1968; Cormen et al. 2022), neither of which ever acts as the expert. A failed attempt consumes no budget: the budget counts demonstrations collected, not prescriptions proposed. After $J_{\max}$ attempts the round falls back to the nearest untried recorded failure. A second screen rejects a prescription the policy already solves. The supplementary specifies it, and no run below exercises it.

Every stage bears on what the policy is trained on and none on how it is fitted, so the evaluation varies the acquisition rule and holds the rest of the loop fixed. The supplementary states the round as pseudocode.

Experiments

Setup

Tasks and policies. We evaluate on five tasks. Each is run twice, with state observations and with image observations, for ten settings in total. GridWorld is a 5×5 discrete grid with three obstacles. Push-T is the ManiSkill3 `PushT-v1` environment (Tao et al. 2024) of the task introduced by Florence et al. (2021); Lift, Wipe and Door are RoboSuite UR5/UR5e tasks (Zhu et al. 2020). Within a setting every method uses the same policy. GridWorld (state) uses an MLP, GridWorld (image) a CNN. The four robot tasks use a diffusion policy (Chi et al. 2023), which takes images through an R3M encoder (Nair et al. 2022). R3M supplies the policy's features and never enters the partition, which is geometric in every run. The expert is a human on GridWorld and a scripted or motion-planned oracle on Lift, Door and Wipe. On Push-T the expert is a PPO policy (Schulman et al. 2017) that is competent in one rotational direction only. The task's constraint graph excludes the configurations the expert cannot serve. No baseline can represent that competence region, so part of DISEIL's Push-T margin may come from avoiding expert failure rather than better allocation.

Budget and measurement. Every method receives $B = 20$ expert demonstrations at $D = 1$ per round and retrains on a shared cadence m, from scratch every round on GridWorld ($m = 1$) and on every fourth acquired demonstration on the robot tasks ($m = 4$). Three rounds in four therefore analyse a policy that has not been refitted. Each still runs the policy on a fresh set of start states, so the failure set and its partition are resampled rather than repeated. We report the held-out success rate: the fraction of episodes solved on start states never trained on. It is a mean $\pm$ standard error, $\mathrm{SE} = \mathrm{std}/\sqrt{n}$, over $n = 9$ GridWorld seeds and $n = 5$ robot seeds. The initial set $|\mathcal{D}_0|$ is set per setting by a behaviour-cloning data-scaling sweep targeting a round-zero success rate near 50 per cent, reported as Init SR in Table 1. That band is where a fixed budget can be spent well or badly: below it the failure set carries no structure, above it it is empty. The ten settings are not ten independent experiments. A task's two settings share the expert, the reward structure and the reset distribution, so the ten rows carry less evidence than ten separate tasks would.

Baselines. The comparison is against the query-efficient members of the DAgger family (Ross, Gordon, and Bagnell 2011): SafeDAgger (Zhang and Cho 2017), DropoutDAgger (Menda, Driggs-Campbell, and Kochenderfer 2017), EnsembleDAgger (Menda, Driggs-Campbell, and Kochenderfer 2019), ThriftyDAgger (Hoque et al. 2021a) and Diff-DAgger (Lee, Kang, and Kuo 2025). All five decide only when to hand control to the expert, each from a different score. None changes what the round's demonstration is spent on. Diff-DAgger's score is the per-step denoising loss, the same signal DISEIL uses to locate a failure; it runs on the robot tasks only. We add Stagger, a control with no decision rule: it spends each round correcting one uniformly chosen recorded failure. Running it on all eight robot settings was beyond the compute available, so it exists on both GridWorld settings and, as ablation A2 below, on two robot settings. One asymmetry favours DISEIL. Its descriptor is read from privileged simulator state, whereas the five baselines score the policy's own output. The descriptor supplies only the partition, and no policy observes it.

Main results

Highest mean in all ten settings. Table 1 gives the final held-out success rate of every method. The margin over the strongest baseline in each setting averages 2.80 percentage points, with a standard deviation of 1.73 and a range from 0.0 to $+5.6$. The strongest baseline changes from row to row. It is Diff-DAgger on Push-T, Wipe and Door, under image and state observations alike. On GridWorld it is ThriftyDAgger for state and Stagger for image. On Lift (state) ThriftyDAgger ties DISEIL; on Lift (image) three baselines are level at 99.6. The result rests on the margin recurring across rows, not on its size in any one row. The standard errors across seeds overlap in several rows, and no single row would support the claim on its own. On GridWorld (image) the random control reaches 89.1 and the four DAgger baselines 88.4 to 88.8. Deciding when to query the expert therefore buys nothing there. DISEIL still gains 2.2 points, in line with the ablation below.

Lift is at a ceiling, and the ceiling hides the result. DISEIL and ThriftyDAgger both reach 100.0$\pm$0.0 on Lift (state), so the row reads as a tie. It is not one. DISEIL is at 100 per cent after the ninth demonstration and ThriftyDAgger only after the seventeenth, so the same endpoint costs eight fewer expert calls. On Lift (image) DISEIL reaches 100 per cent after the seventeenth. A final number cannot express this once both methods have run out of headroom. Lift also begins the budget furthest above the 50 per cent target band, at 67.2 and 66.4. At a ceiling a null result cannot separate a component that does nothing from one whose effect cannot be observed. The ablation programme therefore runs only where headroom remains.

Where the margin opens. Figure 3 tracks success rate against demonstrations acquired. On Push-T the separation opens from about the fifth demonstration and holds, so Table 1 reports a gap present for three quarters of the run. On GridWorld every method rises together and finishes bunched, so the $+2.2$ on GridWorld (image) is among the smallest margins. Wipe (image) has not plateaued. DISEIL and Diff-DAgger are still rising at the twentieth demonstration, so the 3.7-point gap rests on the final values, not on a demonstrated asymptote. A longer budget could close it.

Information gain

Per-demonstration information gain is the policy's per-step loss on each newly acquired demonstration, taken before the policy is retrained on it (Table 2). Diff-DAgger is the reference here. It queries the expert on the very loss the metric measures, so it is the one baseline that could be expected to lead. It does not. DISEIL is higher in all eight settings

Task	Obs	$\lvert\mathcal{D}_0\rvert$	Init SR	Safe	Dropout	Ensemble	Thrifty	Stagger	Diff-DAgger	DISEIL (ours)
GridWorld	state	20	48.9	85.3±0.9	84.9±0.8	86.2±0.7	86.8±0.7	85.7±0.5	–	**92.4±0.4**
GridWorld	image	20	47.0	88.8±0.9	88.4±0.7	88.8±0.9	88.7±0.6	89.1±0.8	–	**91.3±0.6**
Push-T	state	20	46.2	82.0±3.0	84.8±2.7	85.9±2.6	83.2±3.2	–	94.1±2.0	**96.1±1.6**
Push-T	image	20	43.3	78.1±3.5	82.1±3.1	83.2±3.0	79.3±3.6	–	89.0±2.1	**92.6±2.2**
Lift	state	8	67.2	99.2±0.7	99.2±0.4	99.2±0.4	**100.0±0.0**	–	99.2±0.4	**100.0±0.0**
Lift	image	8	66.4	99.6±0.4	97.2±1.6	98.8±0.7	99.6±0.4	–	99.6±0.4	**100.0±0.0**
Wipe	state	12	47.7	88.0±1.1	88.6±1.8	86.8±1.9	89.0±1.1	–	90.4±2.7	**93.1±1.3**
Wipe	image	12	45.2	69.6±2.4	83.2±3.0	84.4±3.2	69.2±4.0	–	88.6±1.4	**92.3±1.4**
Door	state	4	56.8	91.8±2.1	92.5±1.2	88.8±3.1	89.6±1.7	–	93.2±1.9	**96.6±1.9**
Door	image	4	43.1	82.4±1.4	81.8±1.5	83.0±4.9	82.8±1.2	–	84.2±1.6	**88.6±1.5**

Table 1: Final success rate (per cent) on held-out test episodes, meaning start states never trained on. Mean ± standard error over 5 seeds (robot tasks) and 9 seeds (GridWorld), at a budget of 20 expert demonstrations. $|\mathcal{D}_0|$ is the number of initial demonstrations and Init SR the round-0 success rate. The highest mean in each row is in bold, ties bolded jointly. Stagger allocates each round at random; the five DAgger baselines decide when to query the expert. A dash marks a method that does not apply: Diff-DAgger on GridWorld, whose policy is not a diffusion policy, and Stagger where it was not run. The Lift rows sit at the 100 per cent ceiling, so the tie they show is not a tie in sample efficiency. DISEIL reaches 100 per cent after the ninth demonstration on Lift (state) and the seventeenth on Lift (image). ThriftyDAgger needs the seventeenth on Lift (state).

Task	Obs	Diff-DAgger	DISEIL
Push-T	state	1.57±0.49	**2.81±0.93**
Push-T	image	1.80±0.49	**2.82±0.77**
Lift	state	1.61±0.50	**2.64±0.74**
Lift	image	1.36±0.38	**2.93±0.75**
Wipe	state	1.43±0.36	**2.91±0.90**
Wipe	image	1.95±0.52	**3.62±0.98**
Door	state	1.84±0.50	**3.43±0.95**
Door	image	1.58±0.41	**3.00±0.89**

Table 2: Per-demonstration information gain, mean ± standard error over the same seeds: the policy's per-step loss on each newly acquired demonstration, measured before retraining on it, with the higher value in bold. Diff-DAgger is the one baseline that queries the expert on the same per-step loss the metric is computed from, and it is lower in all eight settings in which it runs. GridWorld is omitted because Diff-DAgger does not run there.

in which Diff-DAgger runs, higher than every other DAgger baseline in all ten, and higher than Stagger in the four settings Stagger runs in (supplementary). DISEIL's own target rule (Eq. 7) also selects on peak loss, so the metric is not neutral to DISEIL either. The table establishes only that the acquired demonstrations are not less informative. A high loss before retraining has two possible causes: the demonstration is new to the policy, or it is incoherent. Incoherence is ruled out by construction. An infeasible configuration never reaches the expert (Eq. 8), and the expert's own trajectories are the fitting target. Two readings survive that Table 1 cannot separate: the demonstrations are better, or the budget is better spread across the failures the policy is still producing.

Ablations

The knockouts show that the budget is better spread across failure modes. Each knockout removes one component and retrains, so the drop in success is what that component was worth. Each result is three values, in the order GridWorld (image), Push-T (state), Door (image). They cover a CNN policy, a state diffusion policy and an image diffusion policy. Lift is excluded: it sits at the 100 per cent ceiling, where removing a component can show no drop. The supplementary gives the full programme with margin retained per setting.

Removing the partition costs success without costing information. The most damaging knockout deletes the mode structure but keeps the loss signal, so each round corrects the single highest-loss failure. Success falls by $-2.2/-4.1/-6.8$ points, a mean of 4.37, and the margin retained averages -53.2 per cent. It also drops beneath its own best baseline on Push-T (92.0 against 94.1) and Door (81.8 against 84.2). Per-demonstration information gain does not fall alongside success ($+0.02/+0.16/-0.13$, mean $+0.02$). Picking the worst-loss failure every round buys demonstrations that are informative alone and redundant together. The partition buys coverage of the failure distribution, not a better demonstration. This knockout removes the whole allocation stack: once the modes are gone, the descriptor and the memory have nothing to operate on.

The advantage lies in the allocation. Uniform-random allocation over the recorded failures (A2) reaches $89.1/82.3/80.0$ against DISEIL's $91.3/96.1/88.6$. On both robot settings it lands below the strongest query-gate baseline. Promoting the deterministic fallback rule to the whole method costs 3.27 points. So neither replaying failures nor that rule accounts for the result. The fallback knockout keeps the partition and reverts only the placement, the nearest isolation of *where* available here. It also removes the prescription model, so placement and prescription are not separated. Removing the constraint graph costs 2.37 points and drives the fallback rate to 27 to 35 per cent of rounds. Removing bridging placement costs 1.27. Removing the prescription model, the vision-language model or the cluster memory costs 1.33, 1.33 and 0.73. All three are comparable to the seed standard

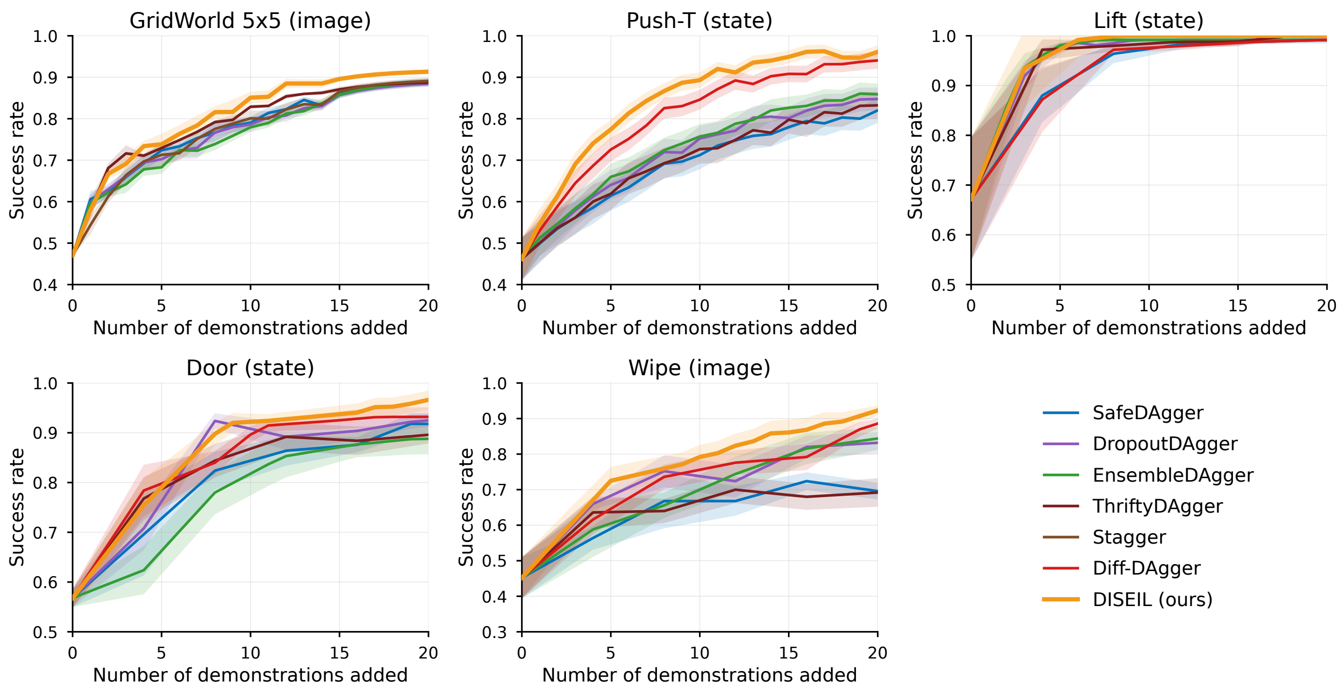

Figure 3: Held-out success rate against demonstrations acquired, mean $\pm$ standard error, on five settings spanning a discrete CNN policy and a continuous diffusion policy.

error of the full run. Replacing the prescription model with a heuristic leaves the Push-T margin at 0.1 points, inside that standard error. A deployment without that model retains the result on two of the three settings only. Dropping the vision-language model as well was not measured. The margin over the baselines is $+9.07$ at $B{=}10$, $+2.87$ at $B{=}20$ and $+2.83$ at $B{=}40$. It narrows because the baselines start lower and catch up. DISEIL at $B{=}10$ still does not reach what the strongest baseline attains at $B{=}20$.

Limitations and Conclusion

DISEIL reasons about one round and holds no representation of the training set. Its only memory records where corrections were placed, not what the data cover. So it cannot tell a failure the set never taught from one it already teaches six times and is merely under-fitted. The descriptor is designed by hand and recovers cause only where configuration determines cause. Root-cause purity per cluster is 0.84 to 0.91, scored against the prescription model's own labels, so it records agreement inside one system. The descriptor is also read from privileged simulator state. So the image results show the allocation transferring to an image policy, not a partition built from pixels. Outside simulation the descriptor would need a pose estimator, whose error would enter the partition. The reasoning stack costs 63 to 1,232 seconds per round more than a query gate, at up to 82,116 tokens. The stack runs only when the next demonstration is chosen, never while the robot acts. No knockout varies *where* with *which* held fixed, so the split of the margin between the two is not measured here. Every expert here is a simulator oracle, which is what lets a budget be counted in demonstrations. Outside simulation the budget is a person's time, and demonstrators are not uniformly correct (Belkhale, Cui, and Sadigh 2023; Brown et al. 2019). The information-gain argument then fails: a high pre-retrain loss identifies new data only if invalid demonstrations are ruled out first.

A gate chooses only when to call the expert. Which failure to correct, and where to place the correction, stay free only while the budget is large. DISEIL makes both. It partitions a round's failures into modes over a geometric descriptor and prescribes a configuration checked for feasibility. It holds the highest mean success rate in every setting reported here. Removing the allocation stack leaves per-demonstration information gain untouched while success falls by a mean of 4.37 points across the three settings ablated: the demonstrations stayed informative and stopped covering the failure distribution. The parts that place the correction each cost between 0.7 and 2.4 points, inside the seed standard error of the full run. So this study does not measure what the prescription is worth on its own. Choice of gate moves the result by as much as 19 points, on Wipe (image), yet no gate exceeds DISEIL in any of the ten settings. What a round buys turns on which failure it corrects, not only on which scalar opens the gate.

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